

Ziegelwerkstrasse

Rue de la Briquetterie

BRICKYARD ROAD

THE Magazine of the International Brick Plant Operator's Forum
and The National Brick Research Center

Strada della Fabbrica di Mattoni

Camino de la Planta de Ladrillo

Vol. 2 • Number 1 • January 2003

**Report on
the 48th
Forum**

**Carolina
Ceramics
Plant**

**New
Chipping
Experiments**

**Plasticity
Tutorial**

**Thermal
Expansion**

**Evaluating
DLA
Reagents**



A New Magazine for Plant People!

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Brickyard Road is a magazine for brick plant personnel. The magazine's name reflects the fact that most of its readers drive down "brickyard road" every day to go to work, and the plant is obviously located at the end of that road.

The goals of the magazine are simple, and each issue may include:

- Practical tutorial articles on brick manufacturing topics.
- Practical articles on use of bricks, standards, and testing.
- A plant visit featuring a new plant or improvements to an existing plant.
- News for the brick industry.
- Information on the future of the industry from a manufacturing perspective.

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On the Cover: Fletcher Steele, President of Pine Hall Brick, Fletcher's son Walt, David Steele, and Richard Bickerstaff chat before cooking steaks at the 48th International Brick Plant Operator's Forum last September 30th. The Steak Cookout was held at the Owen Pavilion of the University Conference Center. The Conference Center and Littlejohn Coliseum will both be used for the hospitality and the meetings respectively in 2003.



Just Down "Brickyard Road" . . .

Denis Brosnan, Editor

Real Brick Men Get Their PSA Checked

I WAS ALONE IN MACON, Mississippi, in Pete Pappas' office at Boral's plant last July when I learned I had prostate cancer. We were just beginning our Summer Meeting's plant tour in Macon when I called my urologist. I sat there in stunned silence after receiving the news. After a while, I joined the plant tour and "skinnied" up close to Richard Bickerstaff. I guess just being around Mr. Richard and knowing he had survived the same problem gave me some form of support.

That evening was perhaps one of the nicest nights of my life. We had a chance to recognize everything J.C. Steele has done for us by presenting them with the Director's Award. We had a little fun with the Mississippi State grads and their Dean, Wayne Bennett.

A lot of harsh realities and a few good things were to follow. One of the best was talking to a lot of guys who had the surgery and are now cancer free. Still, there was the gloom on the horizon knowing we lost John Grogan last year from the disease.

I do not want to bore you with my problem. I can say, in an all-encompassing summary, that the surgery and the recovery were much easier than I expected. If you have to go to an urologist with this problem and go *early enough* to one who knows his/her stuff, you will have the same experience.

The key is early detection using the simple *PSA blood test*. I paid \$25 extra to have PSA included for my annual physical exam. It saved me from an excruciating death.

You should know your PSA number just as you know your cholesterol and your blood pressure numbers. My urologist told me a few things:

- Your chances of prostate cancer are roughly the same percentage as your age. Many doctors start looking at PSA when you reach an age of 40.
- What a doctor should look for is a *change* in the PSA number (from your last test) as well as the absolute magnitude of the number.
- You can beat the cancer only through early detection using the PSA test and other ways I won't describe. The famous "digital exam" is not enough!

I am on my soapbox because I had a heck of a scare and came out of this thing unscathed. I also know brick men don't complain much and work very hard all of the time. There is little time for an annual physical. Perhaps, we view it as "unmanly" to go to the doctor.

Just remember that your *prostate* can leave you *prostrate*. Be a real brick man and check your PSA for yourself, your companion, your kids, or your grandkids.

Just DO IT. ■

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Wisdom From The Chairman

**Terry Schimmel, Chairman
The National Brick Research Center
(Boral Bricks)**

WELL, AS USUAL THERE IS A LOT of activity at your National Brick Research Center. Backing up just a little, I want to bring to your attention that for all practical purposes the pilot plant at the Center is complete and includes processes for material preparation, extrusion, cutting, drying and firing. This includes the ability to make 8" and modular size units both solid and cored. Drying can be accomplished on a single unit basis or any custom brick hack of multiple units. Controls and instrumentation are available to create almost any drying curve as well as monitor weight loss and dimensional changes during drying. Similar capabilities for firing exist to include the burning of test size samples, full size units and full hacks to include flashing. These now fully commissioned capabilities along with complete and comprehensive testing from raw materials to fired products really brings the Center together into a facility we should all be very proud of.

A couple of columns ago, I made an appeal to each of you to support the new suppliers venue at the Forum. My observations were that most vendor displays were well attended and plans are being made to address any shortcomings so that this event will be even better. I don't think anyone will argue with the success of the steak cookout. We should all thank Denis and his staff for the grand effort they put into creating what will become the new tradition for the Forum.

To reflect on the Forum, keep in mind the numbers who attend, where they all come from, the variety and quality of the presentations and the just plain wholesome exchange of information and camaraderie that exists at this event. I know of no other industry event that compares in many of these respects, and we thank you for attending and will be looking for you next year and in the years to come.

On another note, I would like to welcome any readers from the Western states and particularly the Western States Clay Products Association. We are sending this magazine to each of you at no charge in an effort to demonstrate the value the Center has to offer. Needless to say, we would welcome your membership in the Center, and I think we will only be satisfied when we have "100%" membership of the industry.

On a final note, I would like to remind you of the upcoming Center Winter Research Meeting at Clemson on February 4th and 5th. This is an important meeting, and this is the time when we all decide how and on what the Center will spend research money. I hope to see many of you in Clemson this February.

As I say each time: I can be reached at 334-480-2488 or via e-mail at terry.schimmel@boral.com. Denis and I are always ready to hear from the membership be it good or bad. The Center is funded primarily by brick manufacturers. Listening to members and fulfilling the needs of the membership is what we are all about. ■

Identify The Brick Man And Win!



This guy was one of the *best engineers* ever to graduate from Clemson. But wait! He wrote one of the shortest Master's theses ever! Did his father's prominence influence Gil Robinson? We'll never know now. Having also graduated again to bigger and better things a few years ago, he still reads

Brickyard Road. OK, when this picture was taken, the styles in clothing were early "K-Mart". If you know who this is, be the first to supply an answer to brick@clemson.edu to win a valuable Center prize.

Last Issue's Winner: Brenda Coley of the Brick Association of the Carolinas was the first and only correct answer to our last issue's quiz featuring Mitch Wells' photograph from a 1960's college annual. If we had allowed folks at Hanson Brick to participate, we would have received hundreds of e-mails identifying Mitch. Way to go Brenda!

Have An Old Photograph Of One Of Your Friends In The Brick Business? Mail it to Denis Brosnan at P. O. Box 613, Pendleton, SC 29670. If we use it, we'll send you a T-shirt, and we will return the picture in any case. We can accept scanned pictures at 300 dpi resolution or better sent to bdenis@clemson.edu.

In Memoriam: Mil Barranti

Denis Brosnan

Mil Barranti, a long-time participant in the Forum Companion's Program, passed away after a tragic accident during a trip home while returning from this year's meeting. Mil is survived by her husband Bob and her son Chris Bagley - both long-time attendees of the Forum. She is also survived by three other children and seven grandchildren. Mil was a native of the Leesburg Community in Rankin County, MS.



Mil Barranti (left) and Sue DeLoach at the 48th Forum

Mil will be remembered as being "full of life". I spent a fair amount of time talking to Mil about making next year's Companion's Program an even better experience. She was optimistic about our future meetings and gave me plenty of ideas. I know that everyone who attends the Forum joins me in expressing our deepest sympathy to Bob, Chris, and the rest of the family.

New Version of E-Solutions® Simplifies Combustion Calculations

Rod Schutt of General Shale and Denis Brosnan of The National Brick Research Center attended a recent "Advanced Combustion Seminar" offered by Hauck Manufacturing of Lebanon, PA. The Seminar featured an introduction to e-Solutions for Combustion®, a program facilitating combustion calculations and design of combustion systems.

The Seminar is based on the latest 3.0 version of Hauck's software product. e-Solutions for Combustion® is a user-friendly software package developed for use by the company's engineering staff. The package proved to be so popular with Hauck's staff that the company decided to make the product available to the combustion industry at large.

The training seminar began with a "black box" premise. Given a furnace of a particular dimension, fuel, and output requirement, what are the necessary design considerations that will allow the processing goals to be met? According to Denis Brosnan, "the basic content in combustion courses hasn't changed in years. Students were always led from fuel to furnace. The Hauck software allows a thorough approach starting with the furnace requirements - allowing the student to explore a logical design process." Using the e-Solutions software, students were then assisted in developing design criteria, such as:

- *Burner Sizing* – calculate total fuel input needed for the furnace and process requirements that determine burner selection.
- *System Design* – advantages and disadvantages to various air/fuel ratio control methods, optimum furnace zone configurations and burner selection.
- *Piping Design* —basic air/fuel piping design practices, control component selection, fluid mechanics, and orifice meters.
- *Blowers and Sound* —blower sizing, sound pressures at distances, and blower sound levels with silencers.
- *Emissions* – emissions and design decisions that impact emission levels.



Denis Brosnan (Far Left - NBRC) and Rod Schutt (Center - General Shale) Participate in the Hauck Combustion Laboratory

The software makes easy work of the "what if" question. According to attendee Rod Schutt, Manager of Process Research with General Shale Brick, this was "the most beneficial aspect of the seminar. With e-Solutions, you can examine the effect of modifications to a system quickly and easily."

According to Denis Brosnan "e-Solutions is a very significant technical accomplishment. With many brick plants operating with reduced staff, the use of e-Solutions offers a tremendous time savings for operating personnel." Hauck will offer its annual *Ceramic Industry Combustion Seminar*, March 31 and April 1 in 2003. Theme: "Maintaining Your Combustion System." More information is available at www.hauckburner.com.

Carolina Ceramics and Ceric Combine for Versatile New Plant

Compact Plant Has a Big Mission

Jim Frederic

THE LIST OF requirements for the new Carolina Ceramics plant was both long and demanding. First, space was limited since the company is located at the same busy intersection on the north-eastern side of Columbia, SC where it first started its operation in 1939. Second, the product line from the old plant which featured hand setting was extensive in both size and color. Edge setting, flat setting, and good flashed color ranges were key requirements, and finally, according to Michael Borden, President of Carolina Ceramics, “the plant had to be a ‘winner’ as soon as it was started up. We’re a job site plant and brick is our only income,” he said. “We have to make orders when scheduled and provide our customers with the highest quality product in a timely manner.”

The plant construction was a joint effort between Carolina Ceramics and Ceric. The company’s responsibilities, headed up by Vice President of Operations, Bob Bond, were the building, all the concrete, the electrical, and the monorail. Ceric supplied most of the other equipment and supervised the overall installation. Crews were brought in from as far away as Portugal since the busy local economy could not supply the labor force to meet the requirements of the construction schedule. Site preparations were started in the fall of 1999, and the first brick came out of the kiln in July 2000.

The capacity of the new plant is 40 million SBE’s. One of the two tunnel kilns at the old plant was shut down leaving the production of that plant at 16 million SBE’s for a total company capacity of 56 million.

New Grinding Plant in 2003

Additions to the existing grinding system to accommodate the increased tonnage required by the new plant were one screen and several belts. A whole new grinding plant will be built in 2003 to provide for greater efficiency and reduced maintenance. That new plant will be adjacent to the existing facility and will service both the old and new brick plants.

The local shale and kaolin are both mined and hauled in from about 15 miles away. Carolina Ceramics has been in the same kaolin mine since 1960. Other materials such as clays and granite screenings are purchased in small quantities for certain products. The company also grinds all fired waste to make their own grog.

The shale and kaolin are blended by volume (loader buckets) to produce six basic mixes. The blended materials go in a hopper which then feeds them to a Miller hammermill. Discharge from the hammermill goes to a bank of three Diester 5’ x 8’ screens which remove the – 8 mesh material. Overage from the screens goes to a J.C. Steele disintegrator (which Bob Bond says is a 1929 model that is still going strong!) and is then fed back to the screens. All – 8 mesh material is conveyed to three



Entrance to New Plant With “Commerative” Round-Kiln Stack in Foreground

storage silos each with a capacity of 100 tons. One of these silos services the old plant while two service the new plant.

Additives such as chromite, marble dust, and soda ash are added dry to the blended materials on the belts leaving the silos. Loss-in-weight feeders



Plant Overview With Setting Area (at right), Kiln and Dryer (in background), and Packaging Area (at left)

control the addition rates. Manganese is also used, but it is purchased dry and then made into a slurry by Carolina Ceramics. The slurry is then pumped from storage tanks to the rear of the open pug where it is added. A scale which monitors the tonnage on the belt feeding the open pug automatically controls the addition rate.

The six basic mixes plus the additives and plus the coatings all combine to produce a wide variety of colors. Products can also be flat set or edge set and flashed or not flashed. It is no wonder that the versatility and flexibility of the new plant were major concerns during the design phase.

Plant Production Facilities

The belts from the storage silos feed the blended materials with about 13% moisture content into a J.C. Steele 50 open pug mill. The balance of the extrusion water is added at this location. An E. H. Wright Pugmaster controls the total moisture content, normally around 17%. The water is supplied by a unique system designed by Carolina Ceramics. The storage tank is a 3000-gallon septic tank buried in the floor under the pug mill. The water is pumped by a 1-hp submersible pump. Another identical pump is mounted on the tank and automatically comes on if the first one should fail. There are no problems with the water freezing during cold weather, and the horsepower required is less than that used on similar systems. The Carolina Ceramics' system has operated without problems for over two years as of this date.

A conveyor takes the material from the open pug mill to a J.C. Steele 88C feeder. "Green" scrap from the extrusion and setting areas is also conveyed to this same feeder. The belt from the 88C feeder to the J.C. Steele 90 pug sealer crosses over a magnet for catching scrap metal and a scale for controlling the speed of the feeder. The consistent flow of material to the pug sealer keeps the J.C. Steele 90 extruder running at an average rate of 21,000



Extruded Column Passes Through "Slinger" and on to Setting Area

modular/hour. The extruder is also equipped with a quick-change die so as to reduce changeover time.

Only about 10% of the product line has any type of a coating. The coatings that are used are slurries, and they are applied by an OMIS "slinger" or disc sprayer. This unit is commonly used in the tile industry and is very efficient for applying uniform wet coatings.

Once the column leaves the texturing area, it is cut into short slugs which are conveyed to the setting area. There the slugs are pushed up through the cutter wires leaving smoother more formal edges as desired by Carolina Ceramics for their commercial line of brick. The cut brick are then moved across a marshalling area where they can be either stacked or faced. The two-high rows are picked up by an end-grip head which transports them to a wide conveyor belt. This belt carries the brick to the actual setting area where two Fanuc M410 robots pick them up and set them on the kiln car. The robots are 6-axis units and are commissioned to handle 20 different



Setting Area Showing Push-Up Cutter (at left), Inverter/Stacker (center) and Transport Head (at right)

products ranging from modular to 16" triples (16" x 8" x 4"). They can either be flat set or edge set, but different heads are required for each of these operations. After two years of experience, the production crew can normally change from one size to any other in less than one hour.

The standard car count for edge set modular is 6912. There are 24 packs (6 wide by 4 deep) on each car. The packs are 12 courses high and the pattern is "8 over 3". The green cars leave the setting area and enter a tunnel that is 13 cars long. Fans circulate ambient air in this area, but no heat is introduced at this time. Since the production crews work seven days a week, this area is designed to be a true holding room between setting and the dryer.

Both the Ceric dryer and kiln are 275 feet long and hold 22 cars. There are 60 cars in the system at this time, but Carolina Ceramics is building 6 more themselves to help reduce lost time in several key areas. The dryer has only 1 track with 10 recirculation zones. Three of these zones (the first, middle, and last) have automatic controls. There are also 2 humidity sensors that help control the drying rates. These are critical in reducing drying defects particularly in the high kaolin mixes. The dryer supply temperature is 300°F.

The standard schedule for the kiln is 16 cars per day, and it is an index push. The preheat is all side fired with 3 control zones while the rest of the kiln is top fired with six control zones. The main fuel is natural gas and the standby is propane. Some of the combustion air is preheated now. After a scheduled year-end shutdown for modifications, all of the combustion air will be preheated. Set points in the soak zone range from 2080°F to 2180°F based on the product line. Approximately half of the production is flashed with colors ranging from subtle tones to distinct lights and darks. Even with the elevated set points and increased flashing demands, the fuel consumption for the kiln is approximately 950 BTU/pound. All software on the kiln was provided by Ceric, and the kiln supervisor can make changes or adjustments from at home if necessary. There are no kiln firemen assigned to the kiln, but the firemen from the old plant are available at night to make manual changes or to reset controls or valves.

As in many new installations, the exhaust fan on the scrubber is also the exhaust fan for the kiln. The exhaust temperature is 380°F. The scrubber is the dry limestone type and was built by Helmisch. Carolina Ceramics buys dry limestone from Tennessee. This limestone is delivered by tanker trucks and blown directly into the system. There are no outside storage facilities or additional handling requirements. The scrubber uses about 24 tons of limestone per month.

Cars leaving the kiln are automatically transferred down two storage tracks to the packaging department. The unloading equipment was built by Ceric while the monorail was purchased and installed by Carolina Ceramics. A large gripper head picks up 2 full packs off the car (total of 576 brick) and takes them to a wide belt that runs parallel to the monorail. There are 8 packaging stations where the brick are manually graded. A Signode M80 head applies the main plastic straps and a Signode M50 head applies the metal cross strap.

According to Bob Bond, the company felt that the monorail was better suited to their packaging needs at this time than a fully automated system. It gives them the blending and inspection capabilities that are necessary to meet the demands of their product line. There is enough room on the backside of the machine where product from inventory can be brought in and blended with a current production run. This same room can also be used in the future for expansion and to fully automate the packaging process if that option becomes desirable.

Current plant operations require 2 people in each the mining department and the grinding department.

These departments support both plants with the mining personnel working 5 days a week and the grinding personnel working the same schedule as the new plant (7 days a week). Staffing in the new plant consists of 2 in manufactur-



Transport Head Moves Brick and Spaces Them on Belt



Robots set 16"x8"x4" Units (Green Weight = 31 lbs./Each)



Entrance to Dryer (at right) and Exit end of Kiln (at left)

ing, 10 in packaging (includes fork lift operators), and 3 in maintenance. There is also 1 supervisor in mining, one in manufacturing, 1 in packaging, 1 for the kilns, and 1 in maintenance. All personnel in the new plant work 4 days on and 2 days off with an average work day of approximately 9 hours.

New Plant Features

• Grinding Plant

Miller Hammermill
3 Diester Screens
J.C. Steele Disintegrator
3 100-ton Storage Silos

• Extrusion/Setting (Ceric)

J.C. Steele 50 Pug Mill
E. H. Wright & Co. Moisture Controller
Carolina Ceramics' Water System
J.C. Steele 88C Feeder
J.C. Steele 90 Pug Sealer
J.C. Steele 90 Extruder
OMIS Disc Sprayer
2 Fanuc M410-iHW Robots

• Dryer (Ceric)

Length: 275 Feet
1 Track, 22 Cars
10 Recirculation Zones
Exit Temperature: 300°F

• Kiln (Ceric)

Length: 275 Feet
Capacity: 22 Cars
Fuel: Natural Gas/Propane
Temperature: 2080°F to 2180°F

• Packaging (Ceric/Carolina Ceramics)

Conventional Unloading Head (Ceric)
Monorail (Carolina Ceramics)
Signode M80 Head
Signode M50 Head
Hi-Vac Vacuum System (Manual)

• Scrubber

Hellmich, Dry Lime Type

Summary

Walking through the new plant in Columbia leaves no doubt that the goals of Carolina Ceramics have been satisfied. The facility is very compact and orderly, and both the movement and handling of the product have been minimized. Versatility is a major accomplishment since the plant produces 7 sizes in both flat and edge set versions, 4 other products in edge set only, and 2 other products in flat set only (total of 20 "commissions" for the robots). Flexibility has also been achieved with changeover times of less than one hour.

Display panels show that the products from the Ceric plant closely match the old hand set units, and, with 98% recovery, the goal of delivering the customer the highest quality product in a timely manner is within reach. It would appear that Ceric, Carolina Ceramics, and the customers of Carolina Ceramics all have a "winner" on their hands! ■



Unloading Head Moves Fired Brick From Car to Monorail Belt



Monorail and Car Cleaning Area



Michael Borden, Bob Bond, Ken Newell, and Grady Hanson (right to left, Carolina Ceramics) Welcome Visitor Julian Barnes (far left) From Ceram Building Technologies in England.

The 48th Forum Proves Change Is For The Better!

Denis Brosnan

THE 48TH ANNUAL *International Brick Plant Operator's Forum* had much of the old meeting tradition mixed with a lot of new changes designed to make the meeting a more worthwhile value to brick people. A total of 700 people enjoyed the meeting and its associated hospitality functions.

Dr. Chris Przirembel of Clemson University and Dr. Gary Messing of Penn State University, and President of The American Ceramic Society, welcomed brick makers to Clemson.



Denis Brosnan, Program Director, and Chris Przirembel (Clemson)

Leading the way in innovations for last year's meeting, September 29th – October 2nd, were four keynote speakers:

- *Richard Manning*, President, Hanson Brick and Tile,
- *Chris Aubertot*, Ceri, presenting for Henri Lens, Keller HCW GmbH,
- *Alan Baxter*, Baggeridge Brick, and
- *John Peile*, Claypave Pty.



Richard Manning Speaks First



Harold Newman on Clay Pavers After Receiving the "Best Presentation Award for 2001"

The Conference Center Hosts The Official Hospitality Suites

The University Conference Center provided a "high class" place for friends from the industry to see each other in a relaxed setting. It was also a place where vendors could discuss "a little business" with their customers.

Needless to say, there was great apprehension on how brick people would like the new venue. After all, the old courtyard



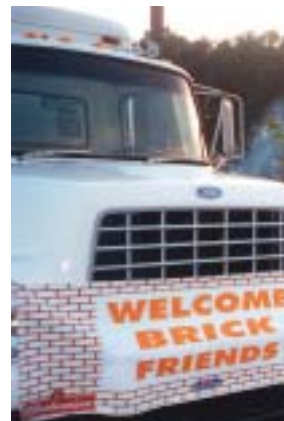
Hanson Men - The J.C. Steele Suite



Signode Always Drew a Crowd

arrangement at the Lake Hartwell Inn was a "certified" Forum tradition.

By the time of the Steak Cookout, everyone was convinced that the new venue was the "place of the future" for the Forum. Now, vendors are working with the Forum Organizing Committee to make sure that the hospitality is even better for 2003.





Mary Ann Keon (Boral), John Steele (J.C. Steele) and Brandon Denmon Discuss the Day's Technical Program



J C Smale's Guys in Front of Their Booth

No Rain But Lots Of Smoke At The Annual Steak Cookout

The new venue for the Steak Cookout was the Owen Pavilion at the Conference Center. Once again, there was no rain for this wonderful event.



ABOVE: Everyone Became a Chef
TOP CENTER: "Can I Giv-A You a Steak?" Says Pete Cieslak of BAC
TOP RIGHT: Ron Davis of Boral is Ready to Cook
LEFT: Budweiser Welcomes Brick Guys
BOTTOM RIGHT: Forum Sponsor Scott McBride on the way to the Grill

Southern Color Hosts The First Brickyard Road Golf Tournament

Southern Color hosted the First Annual *Brickyard Road Golf Tournament* on Sunday. The Brickyard Champion was Bill Tudor of Boral. The 1st Place Team included Bob O'Quinn (General Shale), Mark Bridges (New Riverside Ochre), and Rick Madara and Mike Wyland (McLanahan Corporation).



Ted Corvey, Gary Kenamer, Joel Frasier, and Harold Newman get Ready to tee off

The Companion's group followed their tradition of shopping in Commerce, GA, on Monday of the Forum. On Tuesday, the group went shopping once again with a trip to "The Falls" areas of downtown Greenville with lunch at a local gourmet restaurant. Some ladies ventured out on their own on Tuesday traveling to Highlands, NC.



Sheryle Frederic (left) and Companions Prepare to Visit Greenville



Jim Frederic presents Robby Garland (right) of General Shale the Laptop Computer.

Robby Garland won the laptop computer followed by Jim Bryja having good luck by winning the digital camera. To make a corporate sweep of things, David Strange became the third winner for General Shale accepting a \$100 bill.



Jerry Whitmire (left) Joins the Association Men in the Steak Line

49th Forum To Be Held September 28–October 1

Meeting: 9/29-10/1 at Littlejohn Coliseum, Clemson University.

Hospitality: 9/28-9/30 at The University Conference Center and Martin Inn.

Chipping – Something Old, Something New

Denis Brosnan

Introduction

ONE THING IS FOR SURE – chipping has been around as long as brick have been produced. Interestingly enough, there is no definition of a chip in ASTM C 44, “Standard Terminology of Structural Clay Products” despite the fact that product standards for brick have limitations on the extent of allowable chippage. Perhaps, it is intuitively obvious what a chip is – making a definition unnecessary.

Here’s one anyway:

“*Chip* – area along an edge or corner where material has broken off.”

This definition was found in ASTM F 109, “Standard Terminology Relating to Surface Imperfections in Ceramics”. This Standard has some exciting information. Can you define “kink”, “hole”, or “flow line”? They are all in the Standard.

Perhaps, it is appropriate to find such definitions in standards relating to ceramic products. After all, chippage of hotel china was the original reason people became interested in the subject. In fact, simple pendulum impact tests (Figure 1) were designed in the early 1900’s to test ceramics for chippage resistance. These same pendulum devices have been used with clay brick, but there has been no recent work on the chippage of brick.

Is chippage a continuing problem in brick manufacturing? It depends on your outlook. Plants found over the last few years that addition of rolled edges greatly reduces chippage – particularly at the strapping machine area. In a like manner, chipping of pavers in service is greatly reduced for most brick by adding chamfered or rolled edges. Obviously, these improvements have a fundamental explanation.

Developing Chip Resistance

All of us know that as we fire brick harder, they increase in strength. This is because higher temperatures and, to a lesser extent, longer soaks increase the amount of vitrification (glass formation) within brick. More glass implies more ceramic bond phase, literally “glue”, between residual clay particles. More bond phase generally means more strength.

Most clay brick are, in fact, “particle filled” composite materials on a “micro scale”. Clay particles, packed together during forming, begin to decompose or “dehydroxylate” at

about 925°F. What are left are relics of clay crystals after they have lost their chemically combined water. As temperature is increased above 1475°F, partial melting begins. Melting continues through soak, and vitrified material begins to solidify during cooling.

How much glass is created?

The answer is probably between 5 and 15% by volume – depending on how hard you fired the brick. This means the majority of the brick is residual clay particles. You cannot see them on a reflected light microscope because their index of refraction is the same as that of the glass. Therefore, clay brick on a microscope look like a continuum of glass with occasional pores and islands of quartz and a few other minerals.

It is possible to create too much glass to the point that the brick becomes too brittle. Brick made from alluvial clays – all-clay brick – may exhibit poor chip resistance in comparison to those made from all shale. The reason is that the all-clay brick may be “too well vitrified” as a consequence of its relatively fine particle size.

Where this relates to brick chippage is as follows:

- Chippage resistance should *increase* as “heat work” in the kiln increases up to a point, and
- Chippage resistance may *decline* in the event of excessive vitrification in the brick.

So, given all of this, how do chips form? The answer is that there are two basic mechanisms. In the first place, *impact damage* can create chips. You did not need any reminder of what a mason’s hammer can do to the edge of a brick. As brick become more brittle, they become more susceptible to chippage through impact damage.



Figure 1. Pendulum Tester for Chippage (Courtesy of Acme Brick)



Figure 2. First Chippage Experiment

The second mechanism of chippage is excessive stress near the face of the brick when brick are “squeezed” together. This can happen due to the forces applied to a cube in a strapping machine. The problem is not the strapping machine, but it is the uneven nature of the brick faces leading to stress concentrations at the faces during strapping.

After all, brick have extremely high compressive strength. However, in the compressive strength test the

loaded faces of the brick where loads are applied are “capped” with sulfur or gypsum cement to ensure that an “even” load distribution is achieved on the loaded faces of the brick. This is obviously not the case as compressive forces are applied during strapping.

How We Approached The Problem Of Chippage

Method 1

We decided to measure the force required to create chippage if we *artificially* induced a stress concentration near the face of the brick. We used two 4.72 mm diameter steel rods between two brick – laid parallel to the brick’s faces and near the edge of the brick creating a miniature prism to be subsequently loaded (Figure 2).

The loading was applied at slow rate until 200 lb. of force was reached to fully compress the Celotex pads on the upper and lower loaded faces. Then, the load was applied at a fast deflection rate until a chip was produced.

The method worked, as edge chips were regularly produced. In the case of a “low” compressive strength molded brick (unfrogged), an “initial crack” could be *heard* passing through the brick at a load of about 650 lb. with chips produced on the average at a load of about 3550 lb. In the case of a high compressive strength solid paver, the initial crack was *heard* at a load of about 3325 lb. with chips

produced at about 6000 lb. of load. As anybody who tests brick would expect, there was considerable variation in the data.

We were not satisfied with the method using a two-brick prism separated by the rods. It is important to point out that we were only trying to perform a practical test of a unit’s response. We were not concerned with a scientific property or unit characteristic. The problem was we were using up brick of known physical properties too rapidly. So, we decided to try another approach.

Method 2

In the second approach, we used the rods laid over a steel block on the testing machine platen, with the test brick supported on those rods (Figure 3). We needed the brick to be on “top” of the rods so we could use our deflectometer to measure the traverse of the top platen. This approach worked, and a well-defined chip can be seen in Figure 3.



Figure 3. Second Method to Produce Chippage Using Rods (Deflectometer Not Shown)

We then experimented with 1.6 mm diameter wires (in place of rods) laid in parallel orientation to the brick’s edges in an attempt to produce smaller chips, and we were successful (Figure 4). Of course, this worked since we used reel cutter wires!

The final procedure was to load the brick at a rate of 0.175 inches per minute until failure, or until a maximum

load of 18,000 lb. was reached. The wire on the face of the brick we wished to “chip” was laid as close to the edge of the brick as possible, while the “rear” wire – where we did not want chippage – was laid along the center of the rear face shell.



Figure 4. Chippage Method Using Cutter Wires (Deflectometer In Position)

Results For Method 2 Using Wires

The method produced realistic looking chips (Figures 5 and 6). Chips were generally produced along the edge of the units (Figure 5) – where this particular unit also exhibits fracture across the face. In a minority of cases, chippage was confined to the extreme corner (Figure 6). In a few cases, the unit simply failed in compression without chipping.

The load and deflection diagrams showed either a simple loading to failure (Figure 7) or, more typically, “saw tooth” patterns (Figure 7). The saw tooth patterns corresponded to sequential loss of face chips during the tests.

Data on selected chipping experiments is given in Table 1 along with average physical properties of the units. All of these were well-made units with rolled edges and met criteria for Grade SW facing brick. While there is insufficient data to draw a lot of conclusions, there is an indication that brick with a *higher load* and a *relatively low deflection* at chippage seem to fare better. This, in turn, suggests that brick with a higher modulus of elasticity resist chippage better. Finally, this *could* suggest that brick with *higher compressive strengths* resist chippage better than units with lower compressive strengths.

It is interesting that we were unable to induce chippage in any of the units we tested using the pendulum device shown in Figure 1. Perhaps, the pendulum devices were adequate in a prior era when “commons” were made in addition to facing bricks – sometimes simultaneously in the same round kiln.



Figure 5. Edge Chippage Typical of the Method



Figure 6. Typical Corner Chippage Seen in a Minority of Units

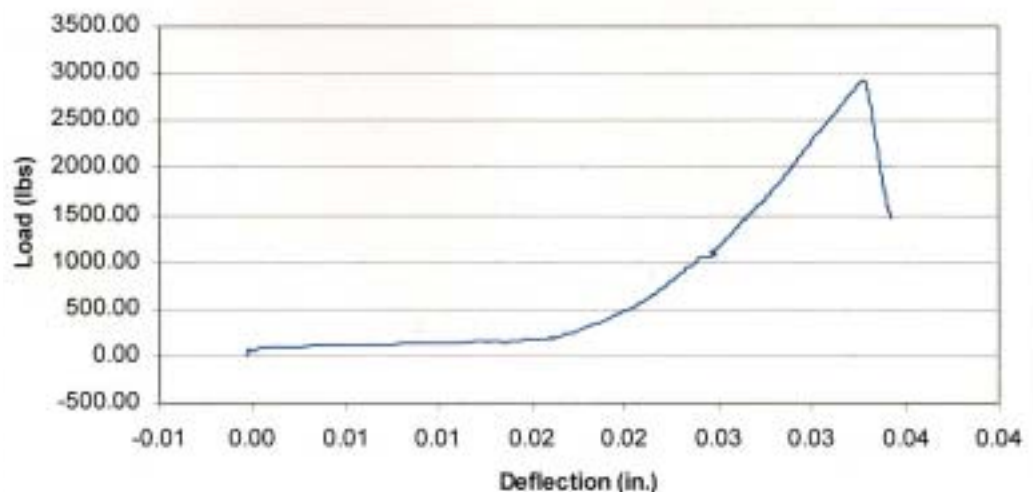


Figure 7. Load Deflection Diagram During Chippage

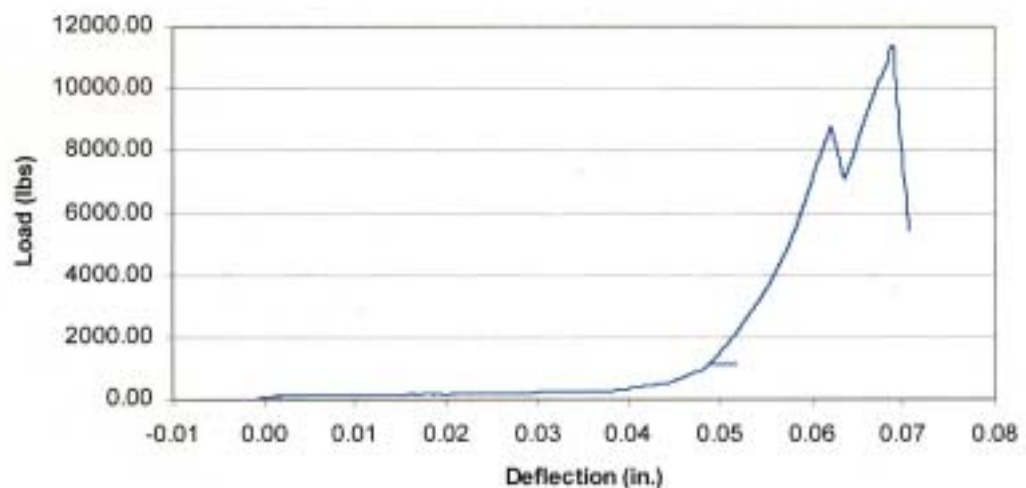


Table 1. Average Chippage Results on Test Brick

Brick	Cold Water Absorption, %	Saturation Coefficient	Average Load at Chippage (lb.) and (σ - lb.)	Average Deflection at Chippage (in.) and (σ - in.)
No. 1, Extruded 3-hole	2.3	0.76	4052 (2511)	0.0747 (0.0244)
No. 2, Extruded 3-hole	2.4	0.58	7391 (6900)	0.0523 (0.0256)
No. 3, Extruded Solid Paver	4.2	0.63	7236 (2869)	0.0586 (0.0123)
No. 4, Molded Unfrogged	4.4	0.93	4089 (1849)	0.0725 (0.0069)

The quantity σ is the standard deviation - in a minimum of five experiments. Data in experiments where deflection exceeded 0.10 inches was discarded.

I would rather build my house or pave my driveway with products exhibiting the least possible coefficient of variation in physical properties. It is not the best – or even the average— product that you make that leads to complaints or scrap. It is the products on the “low end” in the range in product properties that typically fail.

The late Gil Robinson said that coefficient of variation was the “only” statistic a brick maker needed to know. *On this count, the old professor was right.*

Acknowledgement: Thanks to Gary Parker, we were able to develop the method into something that yielded a reasonable simulation of chipping. Stacia Barrow, an undergraduate employee of the Center, performed most of the tests. ■

A Final Word

We never intended to perform a scientific study in this work on chippage of brick. We had a lot of fun trying different testing configurations. Perhaps, we learned what we *already knew* about the importance of high compressive strength in some applications like paving.

We did learn something about variability in data. For example, the coefficient of variation (σ /mean) in load at failure of the No. 3 solid paver is about 43%. This is not too bad for a strength related value in brick. By contrast, the coefficient of variation of the No. 1 and No. 2 extruded facing bricks were 64 and 93% respectively.

How Much Chippage Is Allowed In The Wall?

ASTM C 216 presents information in Paragraph 8.4 on the maximum allowable limits for chippage *in the wall*. The mason bears responsibility to help ensure brick are not placed with excessive chippage. Here are the maximum limits on the “depth” (distance in from the edge or corner) of chips from Table 2 of the Standard.

Type	Percentage Allowed	Chippage in From The Edge in inches	Chippage in From The Corner in inches
FBX	100	1/8	1/4
FBS Smooth	100	1/4	3/8
FBS Rough	100	5/16	1/2

A “small percentage” of units in the wall can have chips of greater “depth”.

Type	Percentage Allowed	Chippage in From The Edge in inches	Chippage in From the corner in inches
FBX	5	1/4	3/8
FBS Smooth	10	5/16	1/2
FBS Rough	15	7/16	3/4

The requirements for FBA are “meeting the designated sample” with requirements not to be more restrictive than those of Type FBS (rough).

Note: Paragraph 8.4 states, “The aggregate length of chips shall not exceed 10% of the perimeter (length) of the face of the brick.” The length is the distance along the edge of the brick that is chipped (ASTM F 109).

Testing Adsorbents for Dry Lime Scrubbers

John Sanders

IN THE APRIL 2002 ISSUE OF *Brickyard Road*, Denis Brosnan wrote an article entitled, "What You Need to Know About Scrubbers if You Have Not Been Working With One" where he described the basics of the operation of Dry Lime Scrubbers. In these scrubbers, a powdered adsorbent such as calcium hydroxide or sodium bicarbonate is injected into the kiln exhaust gas stream. The reagent usually goes through a mixing chamber which allows for good contact between the reagent and the exhaust gas. Acid gases, such as SO_2 and HF, in the exhaust gas are adsorbed by the adsorbent grains to form a new solid compound that can be readily disposed. Ultimately, the reacted adsorbent is removed from the exhaust gas stream by a bag house which uses a fabric filter to remove particulate mater. This type of scrubber is usually used when both SO_2 and HF are present in the gas stream.

This system, like most control systems for acid gases, relies on the reaction between a basic adsorbent and the acid gas to remove the acid gases from the kiln exhaust. Proper adsorbent selection is critical to achieve optimum performance of any flue gas cleaning system. From experience, it is understood that specific adsorbents work well for certain acid gases, but there has been no laboratory scale technique to compare the absorption potential of different types of adsorbents.

A variety of adsorbents are used in dry lime scrubbers depending on the quantity and type of acid gases in the exhaust gas. For hydrogen fluoride (HF) and hydrogen chloride (HCl), hydrated lime or calcium hydroxide ($\text{Ca}(\text{OH})_2$) are typically used as the adsorbent, while for sulfur dioxide (SO_2) sodium bicarbonate (NaHCO_3) or soda ash (Na_2CO_3) are frequently used. Some systems use a mixture of several adsorbents to achieve the required acid gas removal efficiency.

Adsorbent properties such as chemistry, density, porosity, pore size and surface area are understood to be important factors for the performance of reagent materials. The chemistry of the adsorbent influences its tendency to react with certain species. Some typical reactions for SO_2 with adsorbents are given in Denis Brosnan's article in the April 2002 issue of *Brickyard Road*.

How does the adsorption process of acid gases by adsorbent grains take place? How does the reaction proceed?

Basically, the reactions given above take place on the surface of the adsorbent particles as shown in Figure 1. The acid gases react with basic adsorbent on the surface of the grains first and then the acid gas travels into the pores of the adsorbent. After the external surface of the adsorbent has reacted, the acid gases then react with the exposed reagent on the walls of the pores (See Figure 1). In most cases, there is significantly more surface available on the walls of the pores than there is on the outer surface of the adsorbent particles.

The adsorption of acid gases inside the pores of the reagent requires that the acid gas can travel into the pores of the adsorbent. For example, a molecule of sulfur dioxide is many times larger than a molecule of HF, which means that there may be some pores that SO_2 cannot fit into while HF can. Once the available surface on the outside of the particles and the inside of the pore walls has reacted, then the adsorption rate is going to decline significantly.

Further adsorption must take place by a process called diffusion. In this process the acid gas must travel through a skin where adsorption has already taken place and then react with fresh unreacted adsorbent on the inside of the adsorbent grain. Diffusion is not very fast or efficient. Likewise, diffusion is easier for smaller acid gas molecules than for

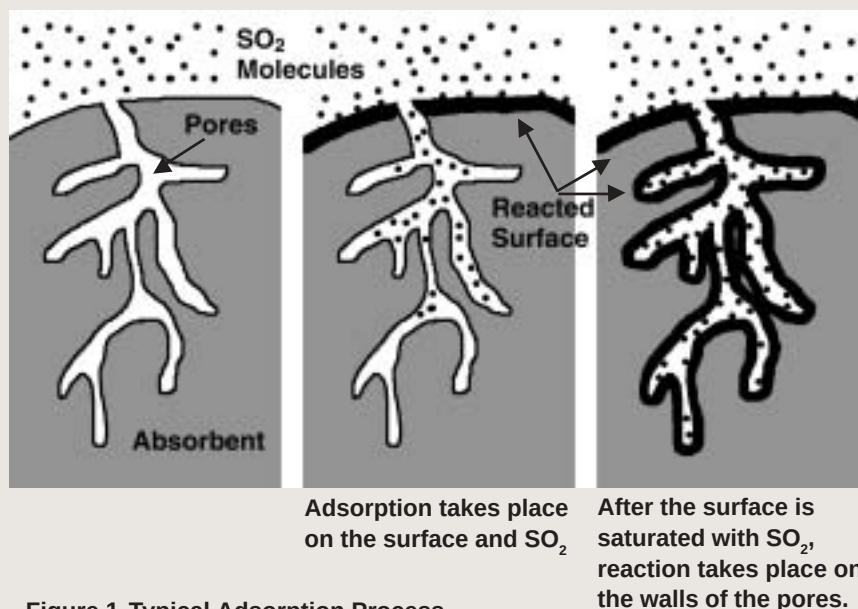


Figure 1. Typical Adsorption Process

larger acid gas molecules such as SO_2 . This means that it is easier for HF to diffuse through the reacted skin and react with the adsorbent than SO_2 . This limitation is the primary reason that powder injection type scrubbers (DLS) are used

to control SO₂ emissions instead of the simpler dry lime absorber. Since DLS's use a powdered reagent as opposed to a granular reagent in a DLA, more adsorbent surface is exposed for reaction.

To evaluate and compare adsorbents for DLS's a new test procedure was developed by the NBRC. Previously, there was no test procedure short of performing a stack test on the scrubber outlet using a series of reagents in the dry lime scrubber, which is obviously too expensive and time consuming to be practical. This new test allows for evaluation of adsorbents in the lab and these results can be taken to the plant. The technique is based on Pulse Thermal Analysis. Pulse Thermal Analysis is a means of calibrating evolved gas analysis signals, but can also be used to investigate gas-solid reactions such as the adsorption of acid gases. A schematic of the test apparatus is given in Figure 2. In this test method, the powdered reagent was placed on the specimen platform in the NBRC's Netzsch STA 449 C. A known volume (1 ml) of SO₂ was injected into the thermal analyzer, and moved over the specimen platform by a carrier gas. The SO₂ was then carried to the FTIR gas analyzer to be measured. When the injected SO₂ flows over the reagent on the specimen platform, some of it passes by and some is adsorbed. When some of the SO₂ is adsorbed, less SO₂ reaches the gas analyzer to be measured, or in other words, when less SO₂ is measured in the gas cell, more adsorption has taken place on the surface of the particles.

For the evaluation of powdered adsorbents, a series of pulses were first measured with no specimen present to establish a baseline peak area. Next, a specimen was placed on the specimen platform and exposed to the same series of

pulses. The amount of SO₂ absorbed was determined by comparing the baseline peaks to the peaks with an adsorbent on the specimen platform.

An example of this analysis for sodium bicarbonate is given in Figure 3. In Figure 3, the SO₂ pulses are given in purple, the weight change of the reagent is given in green and the temperature is given in red. In this analysis, the peaks become gradually larger until they approach the baseline (no sample) peaks. In other words, the beginning pulses display smaller peaks meaning that the absorbent is retaining SO₂. As the absorbent becomes saturated with SO₂, less SO₂ is retained per pulse and the pulses return to their original or empty cell size. Again, smaller peak areas mean that more SO₂ was absorbed. All absorption experiments were conducted at 150°C (~300°F), which is a typical operating temperature for dry lime scrubbers.

There is also additional information present in Figure 3 that is interesting. The reagent sample gained weight with each successive pulse since sodium sulfate, which is being formed by the adsorption of SO₂, is heavier than sodium bicarbonate. Similarly there was a small amount of heat generated when the adsorption reaction took place, which can be seen by the slight increase in sample temperature with each pulse.

For this investigation, five typical adsorbents were selected for evaluation (Table 1). One difficulty encountered with the sodium bicarbonate and sodium sesquicarbonate was their low decomposition temperatures. The decomposition temperatures associated mass losses and changes in surface area are also given in Table 1. Both of these

adsorbents decompose to release water vapor (H₂O) and carbon dioxide (CO₂) at operating temperatures that are typical for a dry lime scrubber.

The decomposition results in an increase in surface area and a decrease in the mass of the absorbent as can be seen in Table 1. These decompositions are also endothermic which means that they consume heat and would tend to cool the exhaust gas as they liberate water. The cooling of the exhaust gas and water generation would result in a potential for condensation and buildup of material in the scrubber when these adsorbents are used. The surface area after decomposition for the calcium hydroxides was not determined since they decompose at higher temperatures than expected in normal scrubber operation.

In a previous study of granular reagents performed at the NBRC, the surface area of an adsorbent has been

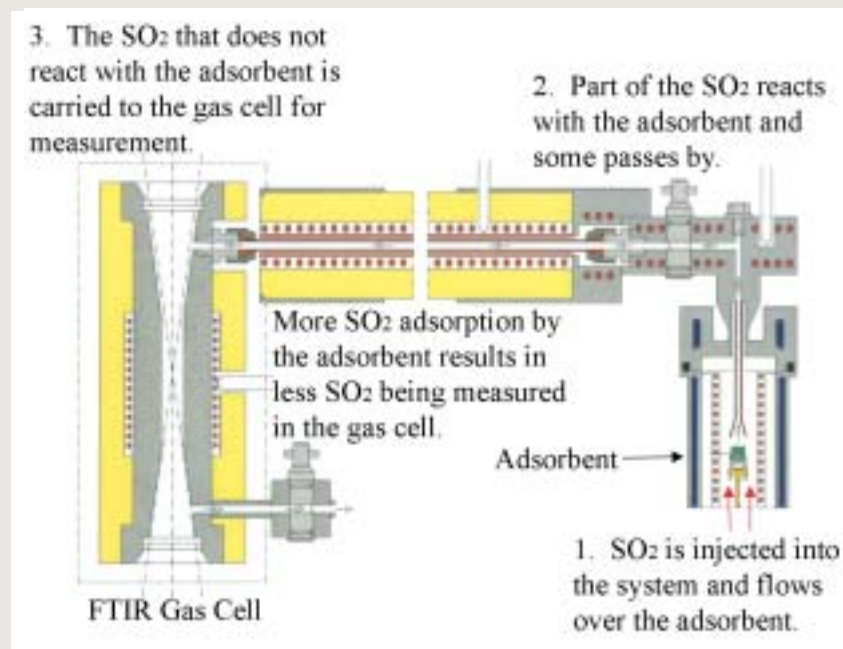


Figure 2. Schematic of Test Apparatus

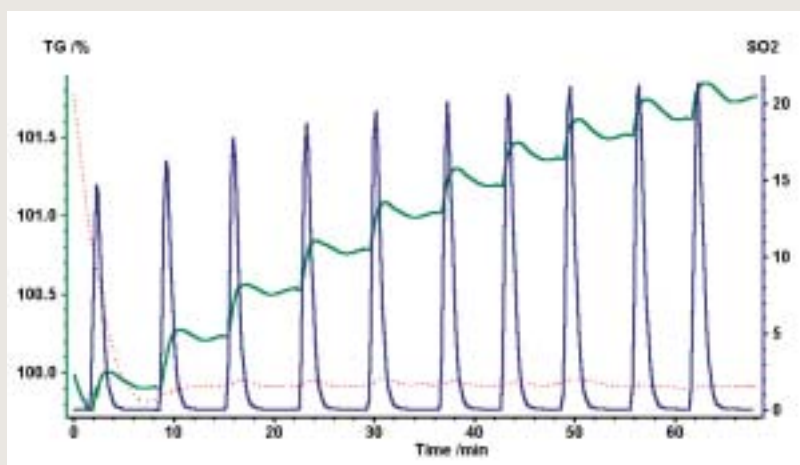


Figure 3. The Pulse TA Method

found to be a very important factor in determining the adsorption potential for an adsorbent. Lastly, specimen B contained approximately 30% MgO which accounts for the relatively small weight loss reported in Table 1. Specimen B is appropriately classed as a hydrated dolomitic lime.

The results of the absorption evaluation of SO₂ by powdered adsorbents are summarized in Table 1 and Figure 4. It should be noted that the absorption values given in Table 1 are reported as the mass of SO₂ absorbed per mass of adsorbent (grams SO₂ / kilogram of adsorbent). Additionally, two absorption potentials are reported in Table 1. The first is based on a raw weight basis as would be analogous to tracking the weight of material consumed by the scrubber. The second absorbed weight is based on the “decomposed reagent” weight for the sodium bicarbonate and sodium sesquicarbonates. For the calcium hydroxide specimens, the absorption potentials based on decomposed weight were not reported.

The sodium bicarbonate (Specimen C) was by far the most efficient absorber. The sodium bicarbonate absorbed SO₂ for more pulses than the other adsorbents meaning that this reagent can absorb more SO₂ before becoming saturated

(See Figure 4). The sodium sesquicarbonate (Specimen D) absorbed less SO₂ than the sodium bicarbonate, but it did tend to absorb SO₂ over the course of a number of pulses like the sodium bicarbonate. For the calcium hydroxide (Specimen A) and hydrated dolomitic lime (Specimen B), higher surface area resulted in greater levels of absorption, but the impure chemistry of Specimen B affected the results. Both of the limes became saturated with SO₂ after only a few pulses (See Figure 4).

In summary, a new test method was developed that allows us to evaluate and compare the adsorption potential of adsorbents for dry lime scrubbers. This test method allows us to compare reagents in the lab so that stack testing is not required at the plant site for a reagent

change. You may be asking yourself, “How can I use this information?” There are several answers to this question, and here are a few. This test can be used to compare the efficiency of different adsorbents so that you can select the reagent that gives you the best chance of being in compliance with your permit. This test can also be used to compare the cost effectiveness of various adsorbents since you may find that with a more costly, but more efficient adsorbent, you can use less reagent and save money (especially important if you also consider disposal costs). What we found in the brief comparison of reagents is that chemistry is by far the most important factor for adsorbent selection for dry lime scrubbers, followed by surface area. Remember your scrubber is only as good as the adsorbent you put in it. ■

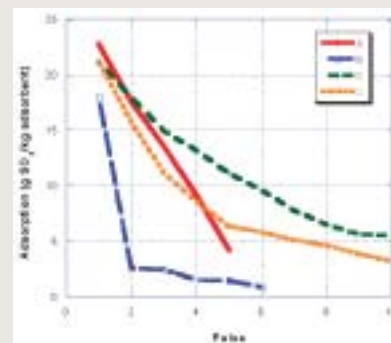


Figure 4. Absorption Trends

Specimen	Type	As Received Surface Area	Decomposed Surface Area (300°C)	Weight Loss	Decomposition Onset	Absorption Potential (Raw Reagent Basis)	Absorption Potential (Decomposed Reagent Basis)
		m ² /g	m ² /g	%	°C	g SO ₂ /kg Absorbent	g SO ₂ /kg Absorbent
A	Calcium Hydroxide	22.0		23.8	383	64.4	
B	Hydrated Dolomitic Lime	11.2		15.5	415	24.4	
C	Sodium Bicarbonate	0.05	1.48	37.0	143	113.5	180.2
D	Sodium Sesquicarbonate	0.35	2.40	29.1	94	85.2	120.2

Table 1. Powdered Reagent Physical Properties



When C Exceeds B In C/B

Denis Brosnan

Masonry Magazine is a wonderful publication with many helpful articles. Therefore, I was surprised, and even shocked, to see the pictures in a February 2002 article entitled "Efflorescence" by Edgar Glock (page 32). The pictures in the article would *imply* that every red brick building ever to be built would have a white efflorescence coating that resembles the mottled appearance of a kid with measles.

"Sand used in mortar is another common source of salts. Sand should be washed with fresh water", concludes Mr. Glock. While this may be sage advice, I wonder if you have tried to wash some sand recently?

Are we going to try this in the old *Whirlpool* you bought when you were first married? How about repeatedly filling and draining a *bucket full of sand* to remove the impurities –

is this sensible at union scale wages? OK – you are brick people, and you think about using a screen and water hose. *Would this be easy?* You know the answer.

In a cool, deliberate moment, I asked my pal Jim Frederic about this. He said you could *always* buy washed sand at a premium price.

What I wondered was why somebody didn't just try the sand before they use it in a Harold Newman like "wicking test". Have we gotten so detail oriented in modern society that we cannot do anything practical anymore?

I have no argument with Mr. Glock. I believe that he must be a talented and competent architect. In fact, I have been impressed with what some architects I associate with in ASTM know about efflorescence. I guess I should give Mr. Glock a heck of a lot of credit – at least he recognized that something besides the brick could cause efflorescence!

Thermal Expansion for the Brickmaker

John Sanders

YOU MAY HAVE HEARD SOMEONE mention the term thermal expansion and wondered what they were talking about. Thermal expansion is simply a measurement of how a material expands or contracts as it is heated or cooled. Materials tend to expand when they are heated and contract when they are cooled, but not all materials or brick, for that matter, expand or contract at the same rate. The thermal expansion of various building materials is usually expressed as a coefficient of thermal expansion, which is a measure of how much a material expands over a certain temperature range.

Thermal expansion can be reversible or irreversible. Reversible thermal expansion means that you can heat it up and cool it down, and it will return to its original size. Conversely with irreversible thermal expansion, the material does not return to its original size; there is some permanent offset. This permanent offset can be due to shrinkage during firing or cracking due to thermal shock to name a few. We will discuss examples of both reversible and irreversible thermal expansion later in this article.

There are several reasons that you might need to know about the thermal expansion of your material. In the brick plant, thermal expansion measurements are used to predict the final size of the brick after they exit the kiln and to determine the optimum firing temperature of the material to get densification (shrinkage). Architects and engineers use thermal expansion measurements when designing buildings. They use the coefficient of thermal expansion to determine where to put expansion joints in a structure (See BIA Tech Note 18 – Volume Changes and Effects of Movement, Part I). Thermal expansion can also be used to identify potential problems in the kiln such as the quartz inversion. Lastly, thermal expansion is also a useful tool for evaluating refractories such as car deck block.

When we talk about thermal expansion, it is important to understand that a material grows in all directions when it is heated, which means we are really talking about volume changes. For convenience, we typically we do not measure the volume change of a material, but the length change and assume that it grows the same amount in all directions, which is typically true for most ceramics. Thermal expansion can be measured in a few different ways. The first way is to heat a material and measure how much the size changed. The coefficient of thermal expansion or CTE is calculated as:

$$\text{Coefficient of Thermal Expansion} = \frac{\frac{(\text{Final Length}) - (\text{Initial Length})}{(\text{Initial Length})}}{(\text{Final Temperature} - \text{Initial Temperature})}$$

The thermal expansion coefficient is usually expressed as in/in/°F or mm/mm/°C or since the length units cancel each other out, /°F or /°C. The problem with this method is that for most ceramic materials the length changes that you are trying to measure are very small and can be difficult to measure repeatably by hand. Additionally, this method assumes that the expansion is constant between the starting and the ending temperature, which is not always true, such as during the quartz inversion.

A second and more common procedure for measuring thermal expansion and the coefficient of thermal expansion is known as dilatometry where a sample is heated in a special apparatus (a dilatometer) that measures the length change as the sample is heated. The measurement of size change is done continuously, as the material is heated, by a very accurate device called a LVDT for linear variable differential transformer. A LVDT is typically capable of measuring length changes to fractions of a micron. An example of a thermal expansion curve measured by dilatometry is given in Figure 1. This curve compares the thermal expansion behavior of a clay/shale fired brick to an all shale fired brick. The thermal expansion coefficient is given for each direction on heating for the temperature range of 100-850°F. The quartz inversion has also been identified in Figure 1 for both brick and as you can see, there is a sudden expansion on heating and a contraction on cooling. For more information on the quartz inversion see Denis Brosnan's article entitled "A Primer on Firing: What's Happening During Preheat?" in the January 2002 issue of *Brickyard Road*.

For the clay/shale brick, the coefficient of thermal expansion (CTE) of $3.60 \times 10^{-6}/^{\circ}\text{F}$ means that for every 1°F change in temperature, the brick grows 3.60×10^{-6} inches for every inch of brick. For example, a 50°F change in temperature, an 8 inch brick would grow 0.0015 inches in the direction perpendicular to extrusion.

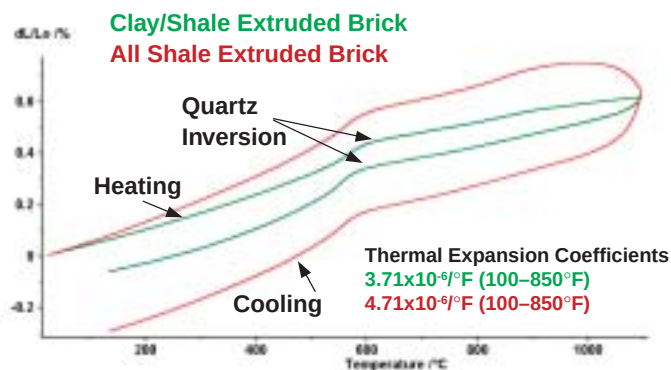


Figure 1. Basic Thermal Expansion Curve

Not all brick expand and contract at the same rate, which means that some brick will have higher coefficients of thermal expansion than others, as seen in Figure 1. For the all shale brick the CTE is higher than for the clay/shale brick. There are several reasons that CTE is different between these two types of brick. Obviously the chemistry of the brick is going to make a difference in the CTE. Brick that have more silica and less alkali content in their composition will have a lower CTE than a brick with more alkali.

The all shale brick in Figure 1 shows more shrinkage when heated than the clay/shale brick and this difference in shrinkage equates to a higher alkali (flux) content. A second factor is how well the brick are fired. Higher adsorption brick will have lower CTE's than brick that have been fired harder. This has to do with the amount of porosity left in the brick. Since the porosity or voids are filled with air they do not contribute to expansion, and in fact they adsorb and dissipate some of the expansion of neighboring material. This effect might also be contributing to the difference in thermal expansion coefficient seen in Figure 1 since all shale brick have a lower water adsorption than the clay/shale brick. Other factors that influence thermal expansion are cracks, laminations, and mineral inclusions to name a few.

Also, remember that the expansion seen in Figure 1 on heating up to about 1600°F is reversible thermal expansion which means that if I stopped heating at this point and cooled the brick off, it would return to its original size. In reality, both brick in Figure 1 undergo irreversible thermal expansion when heated to 2050°F due to shrinkage or densification and do not return to their original size on cooling. In both cases, the brick are smaller after heating.

So far, we have only discussed using thermal expansion measurements on fired brick. Thermal expansion measurements are also very useful for studying how a material changes when it is fired. A comparison of the thermal expansion behavior of the green and fired body for the clay/shale brick is given in Figure 2. As you can see, there is a huge difference between the green and fired thermal expansion.

Basically what you see for the green brick is all the size change that occurs for this body on heating to 2050°F and cooling back to room temperature. On heating, the green body experiences a permanent (irreversible) shrinkage due to dehydroxylation or the decomposition of the clay mineral as well as a reversible change due to the quartz inversion before the onset of shrinkage. The shrinkage that you see for the green body is largely responsible for the development of the final properties of the brick such as adsorption and strength. These properties can be influenced by the amount of shrinkage. For example, if I heated the brick to a higher temperature, I would see more shrinkage in the thermal expansion measurement and a lower water adsorption and higher strength.

For more information on these reactions see Denis Brosnan's article that I mentioned previously and Denis's article in the April 2002 issue of *Brickyard Road* entitled "Chemical and Physical Changes in the Soak Zone of the Kiln". For the fired body in Figure 2, there is a small amount of shrinkage, but since this brick has been previously fired, it has already experienced the bulk of its shrinkage. Thermal expansion measurements on green bodies are used to develop firing curves, compare the firing behavior of different raw materials and identify the sources of certain firing problems.

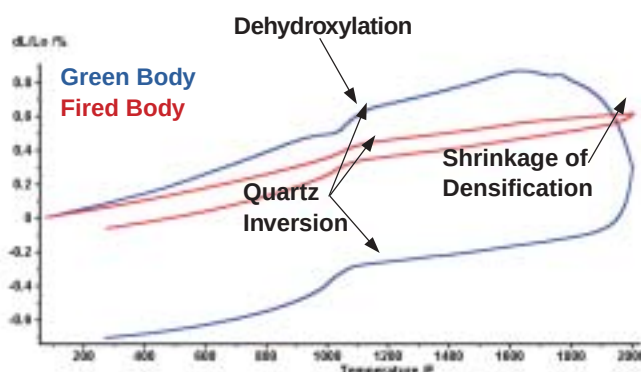


Figure 2. Comparison of Green Versus Fired Thermal Expansion of a Clay/Shale Brick

In closing, thermal expansion measurements are a tool that you can use in your plant to understand what happens to your brick as they are fired. Reactions that take place in the kiln such as the dehydroxylation of the clay mineral, the quartz inversion (on heating and cooling) and the shrinkage of the body can all be observed. For fired brick, the coefficient of thermal expansion is used by architects and engineers to build better buildings with your product. ■

Chips, Cracks, and Bloats

News From the Transfer Pit

Last September 19th, *The New England Journal of Medicine* published an article stating that, "dirt may be actually good for children." Apparently, scientists had a theory about this for about 10 years, and it has been known as the "hygiene hypothesis."

According to the *Greenville (SC) News*, that great bastion of the free press, "farm children in Switzerland, Austria, and Germany – who were exposed to lots of germs (and dust) from animal waste – have lower levels of asthma and other allergenic conditions" (9/19/02). Farm children had 61% less hay fever, 52% less sneezing, and 47% less common asthma than non-farm children."

Since regulatory rationale always centers on "protecting children," we now see that natural dusts may not be the problem made out by the American Lung Association over the last few decades. Neighbors of brick plants need to know this valuable information. It is also tempting to inform our mothers that they did not have to scrub so hard when we were kids!

Environment, Health & Safety News

Information For Brick Plant Personnel To Work Safer And Smarter

Featured Plant Accident:

Casting Failure Puts Loader Cylinder In The Operator's Cab

Prepared By The Brick Industry Association's EH&S Committee

During October of last year, an employee started his shift at 7:00 a.m. The employee performed his normal duties that morning loading clay in the plant shale and clay hoppers. Also, he filled all three special shapes hoppers with nothing out of the ordinary. At 11:30 a.m., the company dump trucks began to bring loads of clay and dumped the clay under the clay shed. At 11:45 a.m., the company dump truck finished his drop off and left.

The operator began to pick up the clay and place it in the bulk storage pile. As the operator proceeded to lower his bucket and enter the clay pile, he had to raise the bucket approximately 8 feet (eye level with operator), from the ground to place the clay on top of the bulk storage pile. The operator had his loader bucket 2 feet above the bulk clay pile and was starting to dump the clay when he heard a loud pop, 10 – 12 seconds later the hydraulic cylinder flew back into the windshield of his Cat front end loader, shattering the windshield. Glass flew all over the operator and the bucket dropped on top of the bulk storage pile. The operator switched the loader off and pulled up his emergency brake and exited the loader and found the Plant Safety Officer, and reported the incident.

Fatality In Demolition Of Round Kilns

An independent contractor with numerous years experience in the brickyard was killed recently when a kiln wall collapsed on him. The project involved removing the steel bands surrounding a round bee hive kiln used by former owners.

The contractor and his assistant cut the bands starting at the top of the kiln and working down. They began work at approximately 10 a.m., took a lunch break, and resumed their activities. At some time around 12:45, there were reports that the kiln had collapsed outwards, trapping the contractor under the rubble. According to his assistant, the contractor was approximately 3 feet from the outside wall of the kiln when it collapsed. He had been cutting up the steel



bands into 8-foot sections, which could then be safely transported by a forklift.

Emergency help (911) was called immediately, and numerous plant employees began to try to remove the brick covering him. He had largely been uncovered when emergency services arrived and took over. He was removed to a stretcher, CPR was conducted, and he was transferred to the hospital where he was pronounced dead.

MSHA Offers MSDS Examples

Examples of MSDS sheets for such materials as limestone, sandstone, natural sand or gravel, and granite are found on MSHA's web site (<http://www.msha.gov/hazcom/hazcom.htm>). These examples provide an "acceptable format" for MSDS sheets for MSHA. As an example, the percent crystalline silica in limestone is given as "Approx. >1%". With respect to silica, the statement for limestone also says "composition varies – typically contains quartz". This *could* be construed as a level of accuracy in reporting as "acceptable" to the agency. ■

Short Courses for 2003

- **Brick and Heavy Clay Manufacturing** (Basic Course) - March 10–14 and August 11–15
- **Fuels, Combustion, and Kiln Operation** October 27–30

Call 864-656-1094 for further information or register online at www.brickandtile.org

Plasticity – Essential Facts for Making Brick

Denis Brosnan

Introduction

IN HIS *DICTIONARY OF CERAMICS*, Dodd tells of two “essential” characteristics for a brick raw material:

- It must possess *plasticity* in order for the brick shape to be formed, and
- It must possess a *broad enough firing range* (in temperature extremes) to avoid too much deformation (melting) at any position on a car in firing in a production kiln.

I am always quick to point out that no particular mineral is specified in Dodd’s definition – whether it is clay, shale, fireclay, etc. In a like manner, no particular limits on chemical analysis are given as essential – although arguably mineralogy and chemistry affect a lot of things including firing range. Also, no particular particle size distribution is denoted — although, once again, particle size certainly affects plasticity a lot and vitrification rate a little.

I am really fond of Dodd’s definition of “brick clay” because it reflects what men and women have been doing for centuries. You find a spot where you want to make something, dig a hole, see if the clay is plastic, and see if you can fire it to a useful absorption with acceptable losses.

Fortunately, exploration has come a long way so that geological maps, or prior brick making history, point to where good materials may be located. This is always followed by core drilling which includes production of plastic formed specimens. Usually, if there is a forming problem, it is known in advance. Occasionally, you get into material in a given pit that is “short” on plasticity. That is when you need help, and that is why I wrote this article.

Plasticity – What Is It?

Plasticity may be defined as follows: *Plasticity* – the property of moist (wet) clay minerals that allows them to be deformed under pressure such that they exhibit a permanent change in shape once the pressure is removed.

In other words, you can form a shape by forcing the clay through a die or into a mold box, and once the shape is formed, the material can be moved (set) without the shape deforming again – if it has sufficient “green strength” or “plastic strength”.

We use a lot of “plastic materials” that are polymers – like PVC – in our everyday lives. They are called “plastics” because the polymer materials are deformed under pressure

and heat to form a subsequently rigid shape. It is no surprise to any of us that PVC pipe is extruded. But these polymeric materials, while retaining the generic name “plastic”, develop flow and cohesion by completely different mechanisms than the plasticity developed in clays with water.

Wet clays differ considerably in their ability to respond to applied force and with respect to the consequences of the application of force. The “*fat*” or “*strong*” clays undergo a large change in shape on the application of a relatively small force. By contrast “*lean*” or “*short*” clays undergo a smaller change in shape for the same applied force, and higher force is required to make them deform to produce something you want. Shapes made from lean or short clays have a much higher chance of cracking on application of high forces, i.e. they do not deform much before fracturing.

All of this implies that there are special characteristics of clays that result in good or poor plasticity. Before we look at these characteristics, or “factors”, let us look at the origin of plasticity.

The Origin of Plasticity

The smallest particles in your mix are called “colloids” – particles less than two microns in largest dimension (1 micron = 0.001 mm). While the larger particles play a role in your product in many ways, it is the colloids that produce plasticity and contribute to flow in forming.

Furthermore, clay particles can ultimately be separated into crystals that are thin and flat with a hexagonal shape. It is convenient to visualize a deck of cards when thinking about clay crystals. They can “lay flat” – one on top of another or they can be scattered everywhere. One thing is for sure, if the cards are “dry”, they have little cohesion— they do not stick together and you can move them around easily.

Now, visualize a deck of *wet* cards. You cannot shuffle them because they are sticking together. The deck has “cohesion” because the water film between the cards is actually “pulling” the cards together. It is the same way with two clay particles with a water layer in between (Figure 1). This is somewhat reminiscent of an “Oreo” cookie or a sandwich.

According to one of the great old “fathers” of ceramic technology, Professor F. H. Norton, this water layer produces plasticity in systems of particles as the surface tension of the water (the force making water form “drops”) pulls the particles together. At the same time, Norton theorized that the particles would “repulse” each other due to the negative charges on the flat “basal” surfaces. Particles in wet clays, according to his theory, would separate by some equilibrium

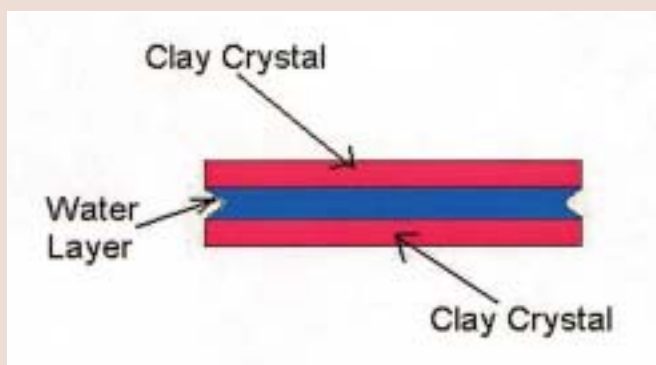


Figure 1: Clay Particles with an Intermediate Water Film

distance based on the water content – more water implies thicker layers in the clay-water-clay sandwich.

How big are these forces between particles, and how thick are the water layers? It has been estimated that the attractive pressures are on the order of 8 kg/cm² or about 113 lb/in². This seems pretty high, but remember that a 2-micron diameter particle has a very small surface area of about $12\text{exp}(-8)$ cm² yielding a force between two particles of this size of about 0.001 grams. This is a small force indeed between any two particles, but amongst millions of particles, the force adds up to provide us with plasticity. Other calculations show that the thickness of the water layer, sometimes called the “*inter-particle spacing*”, is on the order of 0.001 microns or less. This means that the finest clay particles are much larger in dimension than the water layers.

It is when you try to pull the particles apart that you start “stretching” the water layers, until ultimately the particles move and the water layer separates – subsequently forming new “sandwiches” with different particles. When clay is flowing in an extruder, these types of particle-water actions happen repeatedly in numbers of cases that would be mind-boggling. Once the clay particles quit moving (the column is formed), things settle down.

A lot of people have criticized Norton’s theory of plasticity as being too simplistic. It is known, that soluble ions – like potassium – change the apparent plasticity of the materials. Organic matter also changes the situation. This further complexity does not take away from the fact that the simple mental image of an “Oreo” cookie gives us a hint as to where plasticity comes from.

Factors Affecting Plasticity

As plasticity has been described above, it is an *intrinsic* or *self-evident* property of a particular clay. But is this true? The answer is “No”. There is a difference between “Apparent Plasticity” for a raw material at any particular state of preparation and at any moisture content and the “Maximum

Plasticity” that can be achieved under optimum conditions.

Here is what the foregoing means. Simply take a couple of pounds of “shredded” or divided clay at a low moisture content (~1%), and start adding water in a mechanical kitchen-type mixer. Continue adding the water until you reach a level of about 15%. Try two things:

- 1) Measure the plasticity of one specimen immediately.
- 2) Cover the mixer pan with foil and wait overnight. Then measure the plasticity after 24 hours.

Which one was more plastic? If you answered the material after “souring” overnight (#2), you were right. Why is this? The material after a brief period of mixing consists of wet agglomerates dispersed in a “sea” of drier material. When you measure the plasticity, the wet regions have it and the drier ones do not. The overall result is somewhat low “apparent” plasticity.

If you let the material dwell overnight, water will be distributed within the material by capillary suction from wet regions into dry regions. In other words, what you failed to accomplish in the mixer, Mother Nature did for you. Therefore, the material held overnight has a much higher plasticity. The key was water distribution, and people making ceramics have known about this for years.

A Practical Example

At one point in my life, I made refractory mortar similar to the well-known product “Sairset”. I had an old-fashioned ribbon blender that would hold about 1000 gallons of mortar. The poor mixing action (poor efficiency in wetting of the clays) forced me to “over wet” the material knowing that “tomorrow” it would be much stiffer in consistency. The Plant Manager could not understand this, so on the first order he made and packaged the material to an immediate toweling consistency. I watched the truckload go out to a power company one day and come back the next. The mortar was as hard as a rock the next day. The clay particles redistributed the water overnight which had the effect of making the mortar overall less fluid (much stiffer). I understood that wetting is a challenge in development of consistency. It is all in the rate and efficiency of the mixing. The Plant Manager, sadly, had no feeling for the materials he was producing.

These stories reinforce that the amount of water and the efficiency of mixing are crucial in developing plasticity. To reach the optimal or ultimate plasticity of your material, you have to do a real good job of distributing the water or “wetting out” of the fine particles. This is why double pugging has been such a success with materials that have somewhat low plasticity. In fact, “ageing” is on the come-

back. One of the newest plants in the USA wets material before it goes into clay storage. The last extrusion water is added after the material comes in from the reclaimers. Is all of this worth it? Many smart people know that developing the ultimate plasticity before extrusion will pay off in the lowest possible losses!

One quarry tile plant in the USA extrudes their material today for use tomorrow! In other words, they extrude it into a big pile on the floor of the works at the end of the shift, cover it, and re-feed it to the extruder tomorrow. Here is Mother Nature working once again. While I know you cannot go to this extreme in brick manufacturing, I want you to be sure you have absorbed the point – you must use water wisely wetting out particles as efficiently as possible to garner maximum plasticity.

Getting Maximum Plasticity

- Efficient mixing – double pugging.
- Prewetting and “souring” or “ageing”.
- Use of wetting agents and extrusion aids such as “Additive A” and the new colloidal materials.
- Mixing “short clays” with “fat” ones to achieve better overall plasticity.
- Reducing your top screen size to “liberate” more colloids.
- Using steam in the pug mill (an *old* idea).
- Adding a small amount of “bentonite” to the mix.

Plasticity and Water Content

A few things can be said about water. We know it is required to activate plasticity – as a dry brick is not plastic at all. We also know that water is the lubricant between particles during extrusion – allowing particles to slide by one another as the material is compressed into the die and exits the shaper cap. If you do something – like add chemicals – to make the water more viscous, the material will appear to be more plastic (This is the secret of most fast food “milk shakes”).

Above a certain water content (perhaps ~18% in many brick clays), the plasticity declines as the material becomes much softer. Another way of saying this is that there is a range of water contents where plasticity is adequate for extrusion. It turns out that the more colloids that the raw material has, the larger this range in water content is for acceptable extrusion. Remember, too much water implies excessive drying shrinkage downstream resulting in losses.

Effect of Entrained Air

Entrained air due to low vacuum always reduces plasticity. This is because an air bubble just prevents overlap of water coatings on particles from being as good as it can be. Potters

know that air entrained clay appears to be “short”, but that it can recover its plasticity after “wedging” or “de-airing” under pressure. Obviously, good vacuum also has an advantage of producing bricks with the highest possible plastic strength so that handling is no problem.

Ageing and Souring

These arts are somewhat lost in modern ceramic plants. Grimshaw says that the effect of ageing is due to the decomposition of organic matter in the clay that results in formation of organic acids inside of the material contributing to coagulation (flocculation). The result is improved plasticity and uniformity in the material.

Grimshaw even suggests that fungus has a role in souring, and he further says that potters at one time added honey or leftover wine to clay to improve plasticity. I just do not know what *leftover wine* might be.

Pseudo-Plasticity

Anything added to a clay-water mixture can produce improved apparent plasticity, also called *pseudo-plasticity*. While my opinion is that Additive-A type products function primarily as wetting agents, pseudo-plastic materials create a binding action between particles thereby creating pseudo-plasticity.

Here is a list of some materials contributing to pseudo-plasticity in clay-water systems: oils, gelatinous suspensions, hot starch, dextrin, gum, glycerin, various ferments, enzymes, and “biosolids”. All of these are considered “extrusion aids”.

I have extruded completely non-plastic materials (pure oxides) using paraffin wax or water/methyl cellulose mixtures. People who extrude cement board use water and methyl cellulose to develop pseudo-plasticity with Portland cement powder – a completely non-plastic material. ■

Post-Script

In a conversation last May, Pat Brown of Triangle Brick suggested that more information on plasticity could help some plant people. If this article helped you, Pat is the guy to thank.

Friends of the Forum

Lignotech USA/Additive A
Chemical Products Corporation
Reymond Products International, Inc.

Just in Case You Want to Build a Brick Plant on the Moon

Brick Marketing Ideas From The Core Hole Group, Inc. (Pendleton, SC)

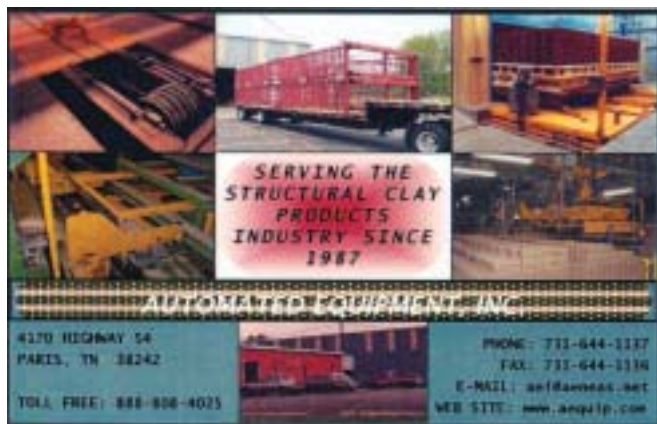
Just when we thought there might be a market to ship bricks to the moon via the space shuttle, our government has sponsored research to demonstrate that bricks can be made using lunar soil. This means somebody will have to build and operate a lunar plant.

In an article entitled "Sintering Bricks On The Moon", Carlton Allen and John McKay say that bricks can be made from lunar material by heating the bricks to 1100°C (2012°F) for two hours followed by *slow cooling*. The article appeared in the *4th International Conference on Engineering, Construction and Operations in Space* (ISBN 0-87262-937-6). Slow cooling – wow— implies some free silica up there? There goes the hypothesis that the moon is made out of cheese.

In a novel bit of research, the authors found that sintering (burning) the bricks in a hydrogen atmosphere produces oxygen (handy on the moon) and produces enough iron metal in the bricks to permit handling of the bricks with a *magnet*. This is something the robotics people here on earth are bound to jump on.

Unfortunately, there is no information available from NASA on mason productivity on the moon. In reduced gravity, a mason ought to be able to reach warp speed. Ever a traveler - *Bill Kjarlien* has offered to go to the moon for trials. NASA is now calculating if their booster is powerful enough to get him into orbit.

So thanks to that gigantic pile of tax dollars in Washington, now we know how to make bricks on the moon. Considering how far a guy could drive a golf ball on the moon, somebody is bound to build a plant there.



Center Winter Meetings

- February 4-5, 2003: Center Offices
Near Clemson (SC)

Center Summer Meetings and Plant Tours

- July 15-17, 2003: Pittsburgh Area (PA)

Call 864-656-1094 for further information.

Vanadium Compounds Can Make You Turn Green

Denis Brosnan

IF YOU EVER SEE GREEN STAINS on fired white colored or buff brick, there is a likelihood that you are looking at “vanadium efflorescence”. Like other forms of efflorescence, the vanadium stains originate through a process of water entering brick that contain soluble salts (compounds) in “rainy periods”, solution of the salts in the water, transport of the water to the surface during drying periods, and deposition of the salts on the surface of the brick. The vanadium salts are typically green in color, but they may also take on brown or yellow tints.

Whenever you see vanadium efflorescence or “stains”, you can be sure that there was a lot of water involved. Places where you can see these stains include garden walls, chimneys, steps, and anywhere else there is a lot of water impingement.

Where do they come from? Recent information shows that the chemical element vanadium in a +4 oxidation state or (V^{+4}) can substitute within the clay crystal for aluminum (Al^{+3}) – the latter being one of the major constituents in kaolin clays. The aluminum – or small amounts of vanadium – are tightly held or – “locked in” the very small clay crystals. There is essentially no soluble vanadium in kaolin as it lies in a clay deposit. In other words, vanadium is not soluble from clays in the ground.

It is only when you fire brick containing kaolin that the vanadium is potentially liberated. This happens as clay crystals “dehydroxylate” or decompose in the preheat section of the kiln – as temperatures exceed about 925°F. Now, the vanadium is “exposed” and it is potentially soluble.

Recent research has shown that vanadium may be slowly *leached* from the kaolin-containing brick, coming to the surface in its +4 oxidation state. When it reaches the surface, it can form complex compounds called *oxychlorides* – compounds of vanadium, oxygen, and chlorine. The vanadium (+4) oxychloride is $VOCl_2$ – listed in chemical handbooks as *green* in color. The vanadium (+4) is thought to oxidize to vanadium (+5) forming another oxychloride, $VOCl_3$ — listed as “*yellow*” in chemical handbooks.

There is a lot of art to cleaning vanadium stains from white bricks, and care is recommended when you may attempt this. Always clean vanadium stains according to your company’s recommendation. It is known that cleaning masonry with muriatic acid (hydrochloric acid) makes the problem *worse* since the vanadium salts in the fired brick are more soluble in this acid.

Nothing is ever quite as simple as the story just told, and other soluble salts in the brick can participate in the efflorescence process. Complex potassium salts, including $KVO(SO_4)_2$, have been identified on the surface of brick. In addition, molybdenum (Mo) may be a part of the complex salts comprising the efflorescence. Finally, much of the vanadium in the brick can be “blocked” from solution by the glass phase in the brick – explaining why process adjustments can minimize or eliminate the potential for vanadium staining.

The late Gil Robinson of Clemson “invented” the use of marble dust, as a mix addition, to reduce vanadium efflorescence in white brick. The idea is to slightly increase the degree of vitrification (glass formation) to retard the efflorescence process. Having said that, most authorities agree that the particle sizing of the marble dust is “tricky”, and the ability to reduce staining is also dependent on things like mixing efficiency and firing schedule. Other ways to retard vanadium are addition of magnesium oxide and/or add certain types of flux to the body.

Lab tests to look at vanadium stains can be very practical. You will likely see it in the C 67 Efflorescence test if your brick have the potential to exhibit stains in service. One good test is the “wicking test” of a white brick laid flat in a pool of water – with evaporation encouraged by blowing air over an exposed surface. This is the test “invented” by Harold Newman. Vanadium compounds are not the only ones that can form yellow/green efflorescence on bricks. To be sure of what you are dealing with, you need to send the Center a stained brick for analysis of the elements on the surface.

What do you tell a homeowner if they ask if the vanadium compounds in the efflorescence pose a health risk? The answer is that vanadium compounds are present in many forms in our society (for example – in metals), and low concentrations do not hurt humans. Throughout the world, occupational exposure limits are set at 0.05 milligrams of vanadium (expressed as V_2O_5) per cubic meter. The small amounts of vanadium in efflorescence are far below this threshold. Humans eat about two milligrams per day of vanadium, and it is thought to reduce cholesterol. Web sites abound promoting the benefits of trace quantities of vanadium (see <http://www.healthy.net/>).

For more information, contact the Center if you are seeing *green*.

For more information:

- “The Chemical Form of Vanadium (IV) in Kaolinite”, A.U. Gehring, I. Fry, J. Luster, and G. Sposito, *Clays and Clay Mineralogy*, Vol. 41, No. 6, 662-667 (1993).
- “Vanadium Staining in Fired Clay Bricks”, D.N. Phillips, *Sixth Conference of the European Ceramic Society*, Vol. 2, p. 381-382, Brighton (June 1999).



YARD TALK

Jim Frederic

SINCE ALL THE ARTICLES THAT ARE WRITTEN for this magazine have to be turned in well ahead of the publication date, it is often hard to keep everything in perspective as far as time is concerned. For instance, here it is early November and I'm trying to write a column that you will be reading in mid-January. By the time this magazine is in your hands, Santa Claus will have already visited you (if you were good!) and each of us will be eagerly awaiting the arrival of the IRS on April 15 (if you were good or bad – it could go either way!).

Being in this situation reminds me of my Army days when I was in charge of an Honor Guard. As the orders were passed down from each level, the arrival time was always adjusted to an earlier hour at each step (especially for funerals!) so as to make sure that we were there when we were supposed to be. It wasn't unusual to arrive at a cemetery at 9:00 a.m. and find out that the funeral was scheduled for 4:00 in the afternoon. Many people find it hard to believe when I tell them that I spent much of my military career sitting on tombstones, but those of you who have been in the service know exactly what I am talking about. The point in all this discussion (yes, there is a point somewhere) is that I want to discuss several topics that took place in October. It may seem like old news to you in January, but at the present "pre-publication" deadline in November it is "hot off the presses".

First of all, each of us here at the NBRC, and I'm sure each of you that read this magazine, is very happy that Denis has his surgery behind him and that the results are so positive. We certainly missed his guidance and his input when he was not here. There is a real team effort at the Center and, when a member of the team is "out of commission", things just seem to be a little out of balance. We notice it even more when the one "out of commission" is our point man for administrative affairs and CU political activities. We were all ready for him to get back – despite the fact that, in this modern age of electronic communications, you sometimes can't tell when someone is gone! Did I mention how hard we all worked during this time? Well, we are just glad to be a family again.

The 2002 IBPOF is behind us and most of the activities went according to schedule. There were a few stumbles along the way, but most everyone seemed to be pleased with the new location and the facilities – especially the steak cookout. Our "super" grill met the challenge of grilling some 700 steaks and there were even some comments about not having to bend over so far to cook (apparently some of us are

getting older!). We met with the vendors in November to listen to their comments and suggestions. Then, it's time to start planning for 2003. It's exciting to think about holding the technical sessions in the coliseum next year and then to plan ahead for the 50th Forum in 2004. If you have any special ideas for the 50th, let us know so that we can make that a memorable event. In the meantime, if you have a BIG party scheduled for your backyard, remember that we "have grill, will travel!" We're pretty cheap, too – all you have to do is feed us and provide beverages.

Speaking of the IBPOF, there have been numerous "inquiries" about the fact that all of the drawing prizes were won by General Shale folks. It would be a real coincidence except for the fact that General Shale sent a lot of people to the Forum (those guys like to "get away"), most of them visited as many hospitality suites as possible (apparently Bob Keyes didn't bring enough "home formula" and they had to circulate in hopes of finding something else), and most of them stayed until the end of the program on Wednesday (they might have won even more if Bob O'Quinn and others had not had an "urgent" meeting at one of the local courses). In short, they did what they had to do to win. Our top-notch security team made sure that all was "fair and square". Hey, if you have any more questions, direct them to Jack Hewitt of Canada Brick - he drew the first badge and started the ball rolling in the General Shale direction! There may be some connection there that we've overlooked.

A few words of appreciation are in order for the guys out in the plants who are quick to act when we need raw materials or brick samples to use in projects or courses. Everybody always responds as soon as asked and the materials are here when we need them. It makes all the planning and scheduling so much easier. Some of those who have helped out recently and deserve our thanks are Tom Moorehead (General Shale, Louisville), Mike Vickers (Acme, Lab), John Bolton and Mike Shealy (Hanson, Ninety Six), Mike Stacey (General Shale, Mooresville), Gary Kennamer (Boral, Augusta), Harold Newman (Pine Hall, Madison), and Stu Weller (Hanson, Columbia). The cooperation of these and others (too many to list) helps the Center complete those programs that are vital in our support of the brick industry.

Finally, (a few serious words) our prayers and support certainly go out to Bob Barranti and Chris Bagley in the loss of Mil Barranti (Bob's wife and Chris's mother). We had the opportunity on several occasions during the Forum to visit with this nice lady, and we were all shocked by the news of her accident a few days later. We also send our regrets again to Mike Foster at the loss of his father. His dad played a major role in developing the meeting that later became the Forum. All of the "brick" family is saddened by these events. ■

H. B. Foster, A "Father" of The Plant Operator's Forum, Passes

Denis Brosnan

Bob Foster and Gil Robinson were the best of friends, and Gil gave credit to Mr. Foster for founding the "brick short course" at North Carolina State University that developed into the International Brick Plant Operator's Forum. Bob's accomplishments were numerous, although he is most recently known as Chairman and CEO of Statesville Brick. Bob Foster passed away on September 30, 2002, at age 87.



Mr. H. B. (Bob) Foster (right)

Mr. Foster was Editor of *Ceramic Industry Magazine* and *Brick and Clay Record* prior to World War II. After service in the U.S. Navy, he worked in the brick industry prior to helping organize the Brick Association of North Carolina (BANC) in 1949. He was the first Executive Director of the BANC – which later became the Brick Association of the Carolinas. He is credited with inventing the brick utility wall and publishing many practical articles on the use of brick. He was a long-serving Chairman of the NC Building Code Council. He moved to Statesville Brick in 1969. Bob Foster received a lifetime achievement award from the Brick Industry Association in 1995.

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Midwestern MEV with screen heating. Cost-effective, end-tensioned units for wet or dry screening. Available with one to five decks in 3'x5', 4'x8' and 5'x10' models.

Fast, easy screen changes

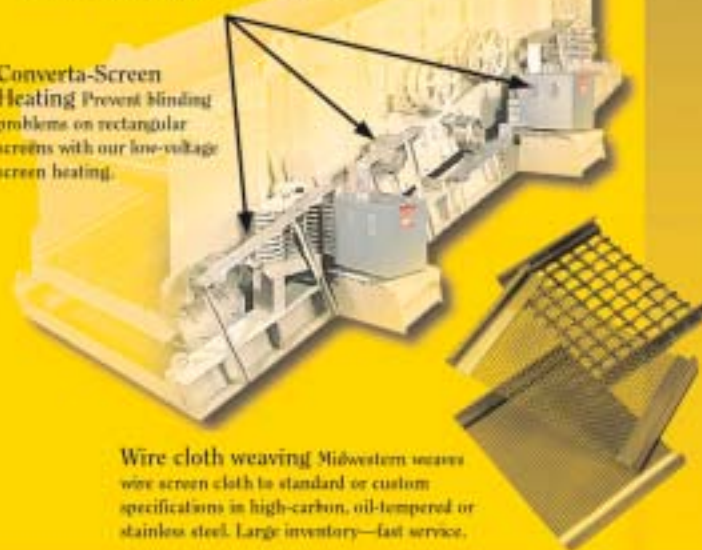


Midwestern Multi-Vib Screen wet or dry material using multiple decks in 3'x5', 4'x8' or 5'x10' models. End-tensioned screens.



The Converta-Screen advantage: low-voltage electrical heating equipment keeps damp material flowing smoothly and efficiently.

Converta-Screen Heating Prevent blinding problems on rectangular screens with our low-voltage screen heating.



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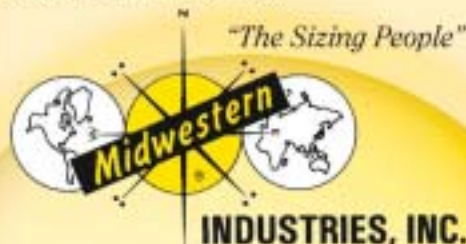
Richtex Brick, Plant 4—Ron Dailey, Superintendent "The MEV does two-and-a-half times the work of our conventional screens. Midwestern's original screen cloth has lasted over 12 months."

Old Virginia Brick Company, Inc.—J. Fletcher Smoak, President & CEO "With the Midwestern MEV, we reduced our work-week from 80 hours to 50 hours. The screen heating on the MEV does a great job—now we have no blinding at all."

Kentwood Brick & Tile Manufacturing—Seavy Clemons, Secretary/Treasurer "The Midwestern MEV increased production 33% and we've changed screens only once. Midwestern's screen heating works fine for us, even with our wet clay."

Boral Bricks (Van Wyck, SC)—Mike Kuvinka, Plant Manager "The MEV has met our expectations and more. Midwestern's MEV produces twice as much as my other screens."

Hanson Brick and Tile—Mitch Wells, Vice President Technical Services "We replaced existing screens with Midwestern screens in four plants. In every case, our tonnage significantly increased and we experienced longer screen life."



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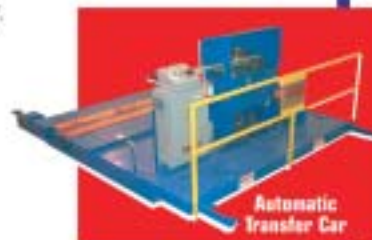
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